

SC Solenoid Field Mapping

New SC Solenoid for Moller Target

INSTRUMENTATION
FEEDTHROUGH

LIFTING LUGS (4X)

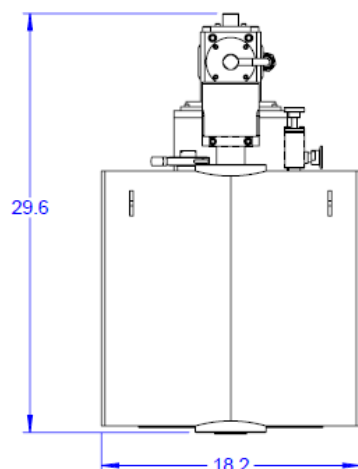
EVACUATION VALVE

CURRENT LEAD PAIR

RADIAL ACCESS

B.C. FOR INTERFACE
PER CUSTOMER SPEC.

- Coming with power supply
- Does not need any cryo lines



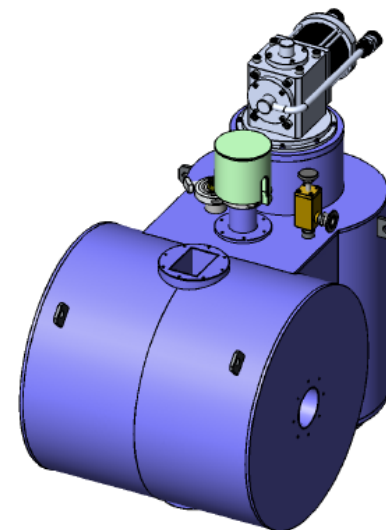
GM CRYOCOOLER

B.C. FOR INTERFACE
PER CUSTOMER SPEC.

Ø 3.00
WARM BORE

Ø 18.0

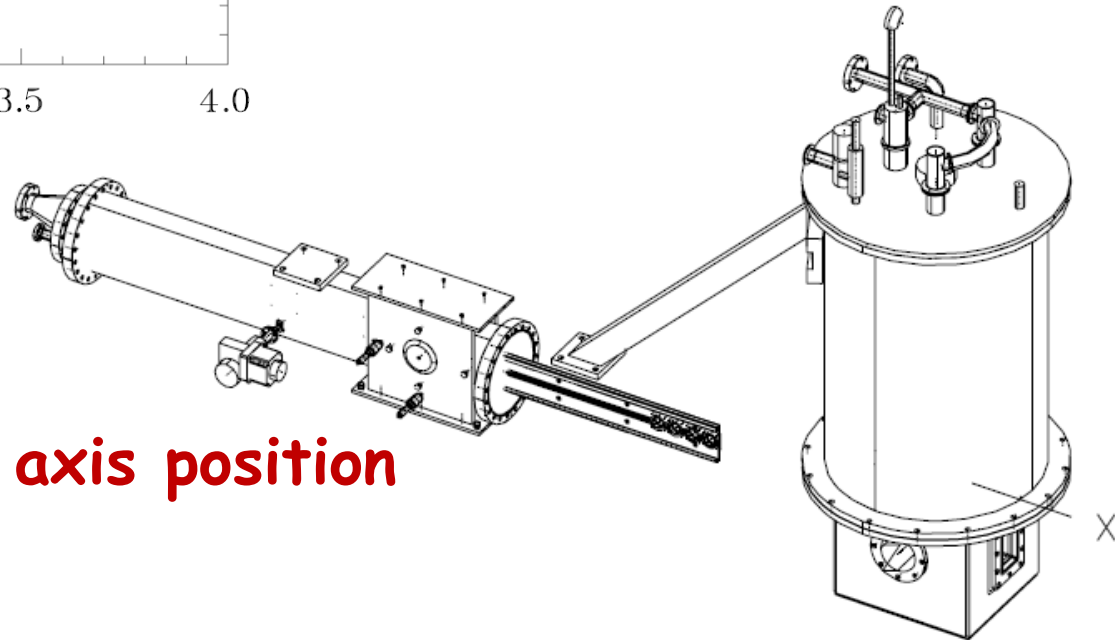
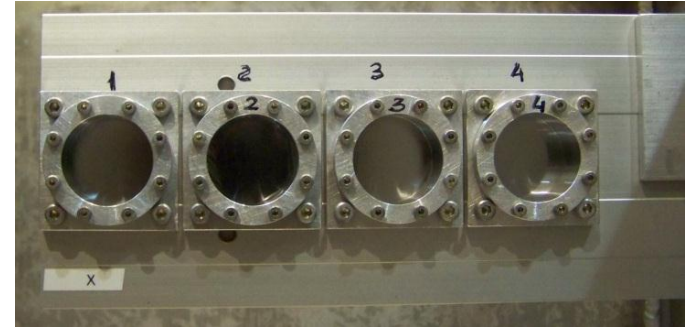
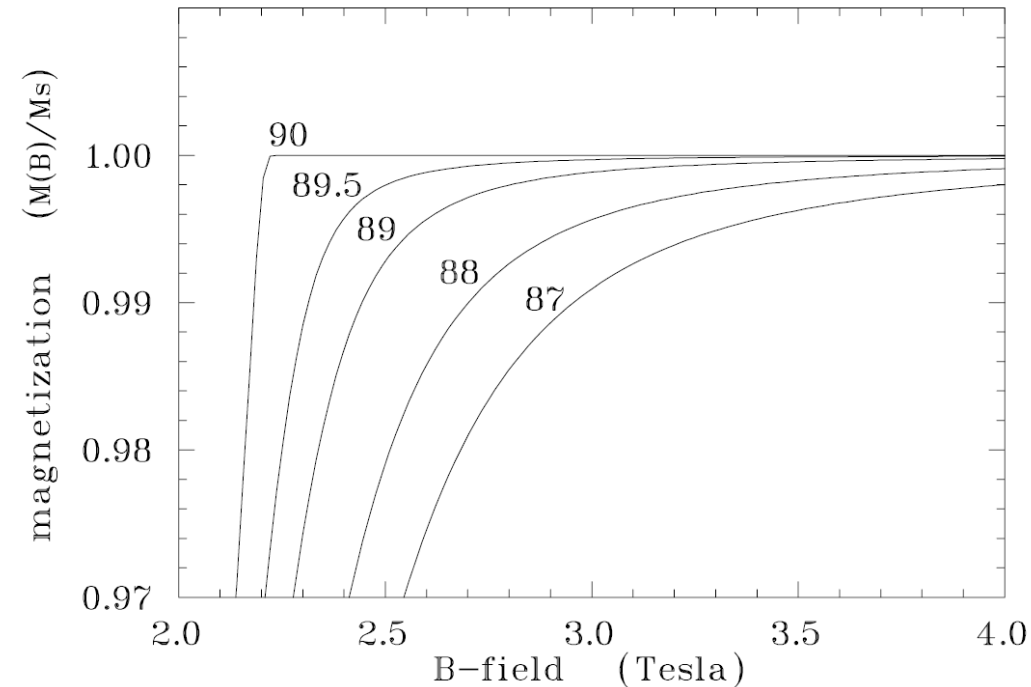
ACCESS PLATE



REVISIONS			
REV.	DESCRIPTION	DATE	APPROVED
A	PRELIMINARY PRINT - NOT RELEASED TO MANUFACTURING		

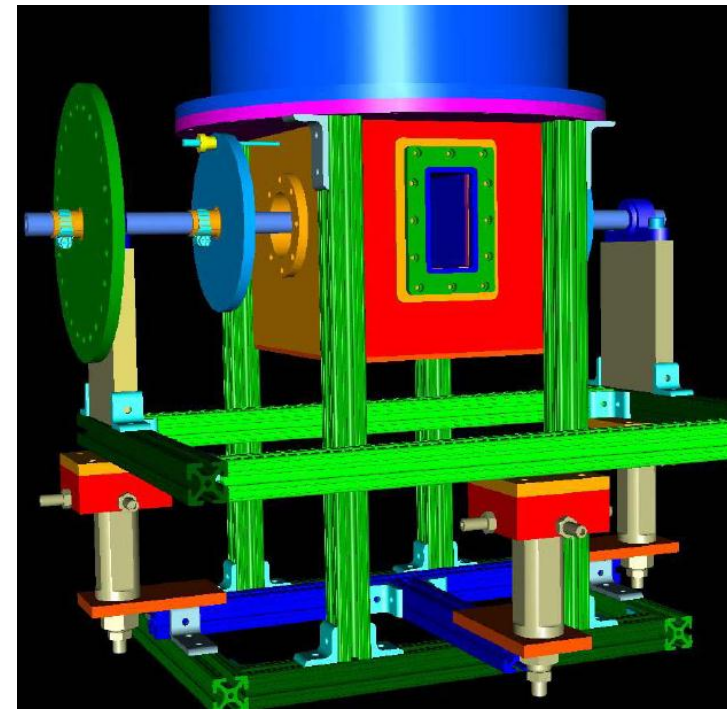
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES TOLERANCES - UNLESS OTHERWISE NOTED:			INSTR. BY: J. AMERICA		AMERICAN MAGNETICS INC.	SERIAL NO.: M21282
ASSEMBLY						
X.X ± 0.1		X.X ± 0.03		JLAB		ST S/C MAGNET WITH RADIAL ACCESS CONDUCTION-COOLED CRYOSTAT
X.XX ± 0.05		X.XXX ± 0.005		DRAWING: A15 6/14/2014		
ANGLES ± 1.5°		ANGLES ± 0.5°		ENG/CNC:		
FRACTIONS 1/8		FINISH		APPROVED:		
REMOVE ALL SHARP EDGES & BURRS				APPROVED:		
PROPRIETARY INFORMATION						
THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF AMERICAN MAGNETICS, INC. OR A SUBSIDIARY THEREOF. ANY REPRODUCTION IN PART OR WHOLE WITHOUT THE WRITTEN PERMISSION OF AMERICAN MAGNETICS, INC. IS STRICTLY PROHIBITED.						
SIZE: B		DWG. NO. SKM21282		REV. A		SHEET 1 OF 1
SCALE: NEXT 1:3						

What Has to be Done

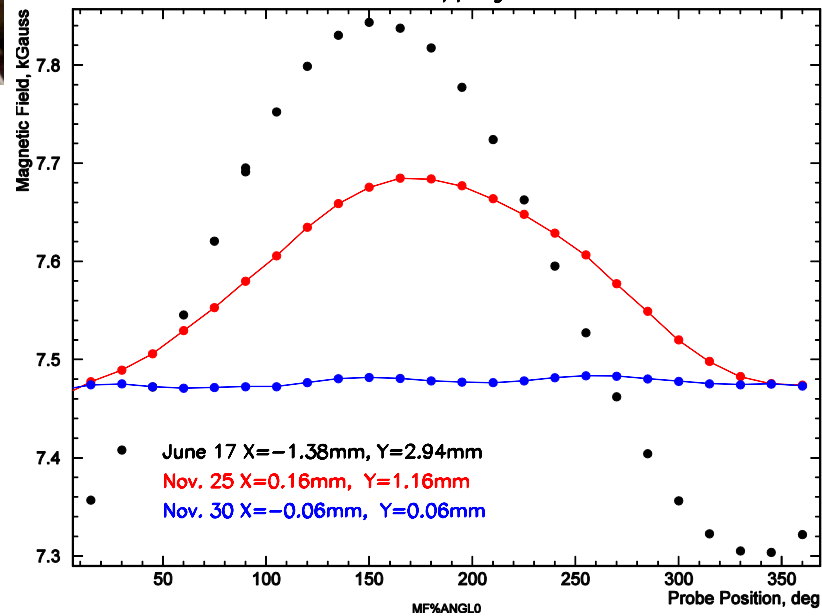


- Precise magnetic field axis position
- Field along the axis
- Field vs. Current

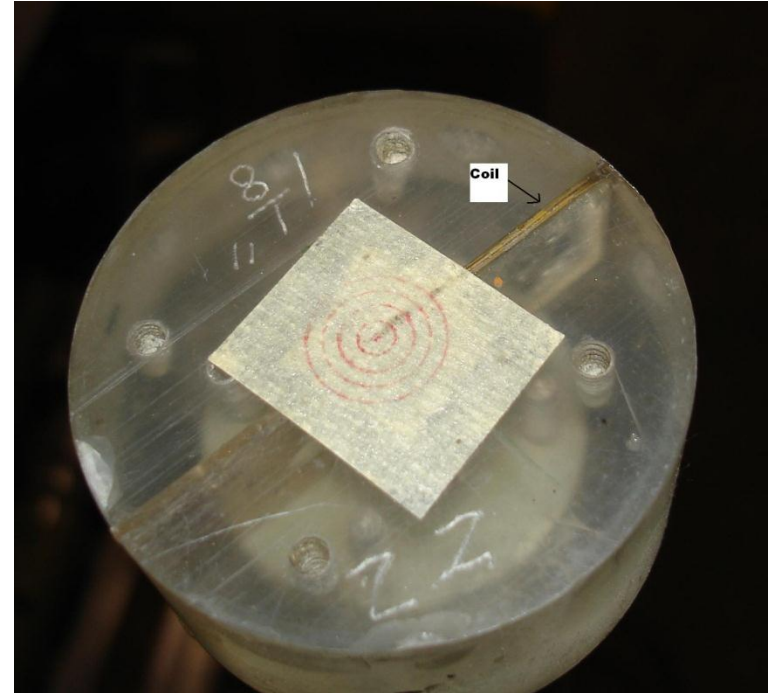
Our Experience



Field Mapping Results



SLAC Experience (from Ken)



For solenoids, we use a short rotating coil on a long rotating shaft (see pict) to do the alignment and measure the dipole and first few harmonics. Our coils rotate at between 10-25 Hz and their signals are readout with a lock-in amplifier (Stanford Research SR850) that is triggered by a pulse coming from the rotation setup (an encoder). The lock-in reads the voltage from the coil for each harmonic and the B field is related to the voltage by $B = V/(r \cdot w \cdot L \cdot N)$, where r is the radius of the coil, w is the rotational frequency ($2 \cdot \pi \cdot \text{speed in Hz}$), L is the length of the coil, N is the number of turns in the coil. Since r and L are not always known for a hand wound coil, we calibrate the coil constant K ($K = 1/(r \cdot w \cdot L \cdot N)$) by putting it in a dipole (uniform) field, B , which we measure using a hall probe and then solving for K using the voltage from the lock in. To align the solenoid in X and Y, one puts the coil at one end of the solenoid where the field is bending, at the pole piece usually, and centers the coil until the signal on the lock-in for the 1st harmonic (dipole) is zero. This is the center for that end. Do the same thing on the other end and find the other center. The center of the rotational axis of the coil is where the center is located. Our coils have targets on the front face that our alignment people can read. (see picture Short Coil with text)